# Homework 3

Solutions

1. (Markov's and Chebyshev's inequalities are tight)

   (a) Consider a random variable $\tilde{a}$ that is equal to $c$ with probability $\theta$, and equal to zero otherwise. Then $\mathrm{P}\left(\tilde{a} \geq c\right) = \theta$ and

   $$\begin{aligned} \mathrm{E}\left[\tilde{a}\right] &= cP\left(\tilde{a} = c\right) && (1)\\ &= c\theta. && (2) \end{aligned}$$

   (b) Consider a random variable $\tilde{b}$ with pmf $p_{\tilde{b}}$ such that

   $$\begin{aligned} p_{\tilde{b}}(\mu - c) &= \frac{\theta}{2}, && (3)\\ p_{\tilde{b}}(\mu) &= 1 - \theta, && (4)\\ p_{\tilde{b}}(\mu + c) &= \frac{\theta}{2}. && (5) \end{aligned}$$

   Then $\mathrm{P}\left(\left|\tilde{b} - \mu\right| \geq c\right) = \theta$,

   $$\begin{aligned} \mathrm{E}[\tilde{b}] &= \frac{\theta}{2}(\mu - c) + \frac{\theta}{2}(\mu + c) + (1 - \theta)\mu && (6)\\ &= \mu, && (7) \end{aligned}$$

   and

   $$\begin{aligned} \mathrm{Var}[\tilde{b}] &= \mathrm{E}\left[(\tilde{b} - \mu)^2\right] && (8)\\ &= c^2\mathrm{P}\left(\tilde{a} = \mu + c\right) + c^2\mathrm{P}\left(\tilde{a} = \mu - c\right) && (9)\\ &= c^2\theta. && (10) \end{aligned}$$

2. (Online poll)

   (a) The proportion of voters in the whole population that votes Democratic is given by $\theta_{\text{tot}} = \alpha\theta_1 + (1 - \alpha)\theta_2$. The fraction of young voters in the poll that intend to vote Democrat $y/n_1$ is a reasonable estimate for $\theta_1$. The fraction of old voters in the poll that intend to vote Democrat $o/n_2$ is a reasonable estimate for $\theta_2$. The estimator for $\theta_{\text{tot}} = \alpha\theta_1 + (1 - \alpha)\theta_2$ therefore equals

   $$e(o, y) := \frac{\alpha y}{n_1} + \frac{(1 - \alpha)o}{n_2}. \tag{11}$$

   (b) We have $y = 50$, $n_1 = 70$, $o = 10$, $n_2 = 30$, $\alpha = 0.25$, so

   $$e(10, 50) = \frac{1}{4} \cdot \frac{50}{70} + \frac{3}{4} \cdot \frac{10}{30} = 42.9\%. \tag{12}$$

(c) Let us assume that the young people in the population are sampled independently and uniformly at random with replacement, and so are the old people. Let $\tilde{y}$ denote the number of young voters in the poll that intend to vote Democrat, and let $\tilde{o}$ denote the number of old voters in the poll that intend to vote Democrat. By the notes, $\tilde{y}/n_1$ and $\tilde{o}/n_2$ are unbiased estimators of $\theta_1$ and $\theta_2$ respectively. Consequently, by linearity of expectation

$$\begin{aligned} \mathrm{E}\left[e(\tilde{o}, \tilde{y})\right] &:= \mathrm{E}\left[\frac{\alpha\tilde{y}}{n_1} + \frac{(1-\alpha)\tilde{o}}{n_2}\right] && (13) \\ &= \frac{\alpha \mathrm{E}\left[\tilde{y}\right]}{n_1} + \frac{(1-\alpha)\mathrm{E}\left[\tilde{o}\right]}{n_2} && (14) \\ &= \alpha\theta_1 + (1-\alpha)\theta_2 = \theta_{\mathrm{tot}}, && (15) \end{aligned}$$

so the estimator is unbiased (under our assumptions).

(d) Assuming independent and uniform random sampling with replacement, by the results in the notes,

$$\begin{aligned} \mathrm{Var}\left[\frac{\tilde{y}}{n_1}\right] &= \frac{\theta_1(1-\theta_1)}{n_1}, && (16) \\ \mathrm{Var}\left[\frac{\tilde{o}}{n_2}\right] &= \frac{\theta_2(1-\theta_2)}{n_2}. && (17) \end{aligned}$$

Assuming $\tilde{y}$ and $\tilde{o}$ are independent, the variance of our estimator is

$$\begin{aligned} \mathrm{Var}\left[e(\tilde{o}, \tilde{y})\right] &= \mathrm{Var}\left[\frac{\alpha\tilde{y}}{n_1} + \frac{(1-\alpha)\tilde{o}}{n_2}\right] && (18) \\ &= \alpha^2 \mathrm{Var}\left[\frac{\tilde{y}}{n_1}\right] + (1-\alpha)^2 \mathrm{Var}\left[\frac{\tilde{o}}{n_2}\right] && (19) \\ &= \frac{\alpha^2\theta_1(1-\theta_1)}{n_1} + \frac{(1-\alpha)^2\theta_2(1-\theta_2)}{n_2}. && (20) \end{aligned}$$

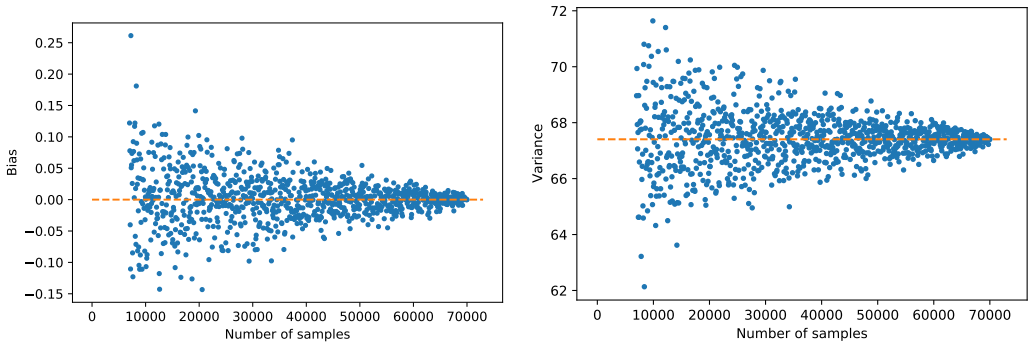
Since the estimator is unbiased, by Chebyshev's inequality, for any $\epsilon$

$$\mathrm{P}\left(|e(\tilde{o}, \tilde{y}) - \theta_{\mathrm{tot}}| > \epsilon\right) \leq \frac{\mathrm{Var}\left[e(\tilde{o}, \tilde{y})\right]}{\epsilon^2}, \quad (21)$$

which converges to zero as $n_1 \to \infty$ and $n_2 \to \infty$, so the estimator converges to $\theta_{\mathrm{tot}}$ in probability and is therefore consistent.

3. (Blood pressure)

(a) The bias converges to 0 and sample variance converges to the population variance.




(b) The following plot compares empirical probabilities with the Chebyshev bound.

